Part 9 – Creating a Client Workstation, Joining the Enterprise Domain, and signing in with an Enterprise User

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Technologies Used

- Oracle VirtualBox
- Windows Server 2022
- Windows 10 Pro
- Active Directory Domain Services
- Domain Name System (DNS)
- Dynamic Host Configuration Protocol

Objective

The objective of this portion of the lab was to create and configure a Windows client workstation that could communicate with the Domain Controller, receive network configuration from the DHCP server, join the corp.corson.com domain, and authenticate using one of the enterprise user accounts previously created in Active Directory.

This portion also allowed me to verify that the Active Directory, DNS, DHCP, and user configurations completed during the previous portions of the lab were working together correctly.

Lab Setup:

This lab continues from the previous Active Directory configurations where Organizational Units, enterprise users, security groups, and permissions were created within the corp.corson.com domain.

A Windows 10 Pro virtual machine named CLIENT1 will be used as an end user workstation. CLIENT1 will connect to the internal network managed by the Domain Controller and receive its network configuration through the DHCP scope configured on DC1.

Once the connection to the domain is verified, CLIENT1 will be joined to the corp.corson.com domain and an enterprise user account will be used to authenticate to the workstation.

Figure 1. Creating the Client Workstation

A new virtual machine named CLIENT1 is created within Oracle VirtualBox and Windows 10 Pro is selected as the client operating system.

Windows 10 Pro provides the capability to join the workstation to an Active Directory domain and will represent an end-user computer within the corp.corson.com enterprise environment.

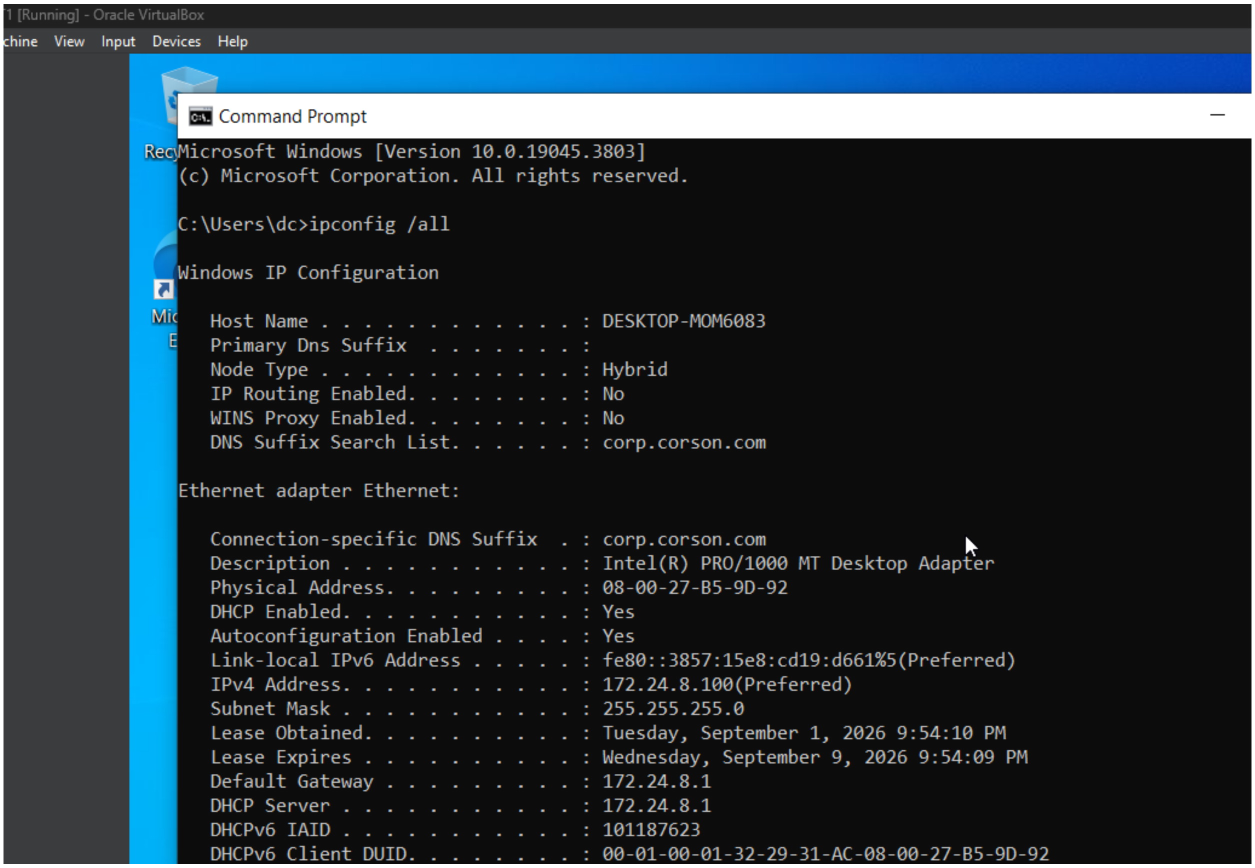

Figure 2. Connecting CLIENT1 to the Domain Network

The ipconfig /all command is used on CLIENT1 to verify that the workstation received its network configuration from the DHCP service configured on the Domain Controller.

CLIENT1 received the IPv4 address 172.24.8.100 with a subnet mask of 255.255.255.0. The Default Gateway and DHCP Server are shown as 172.24.8.1, confirming that the workstation is communicating with the internal domain network.

The corp.corson.com DNS suffix is also assigned to the client.

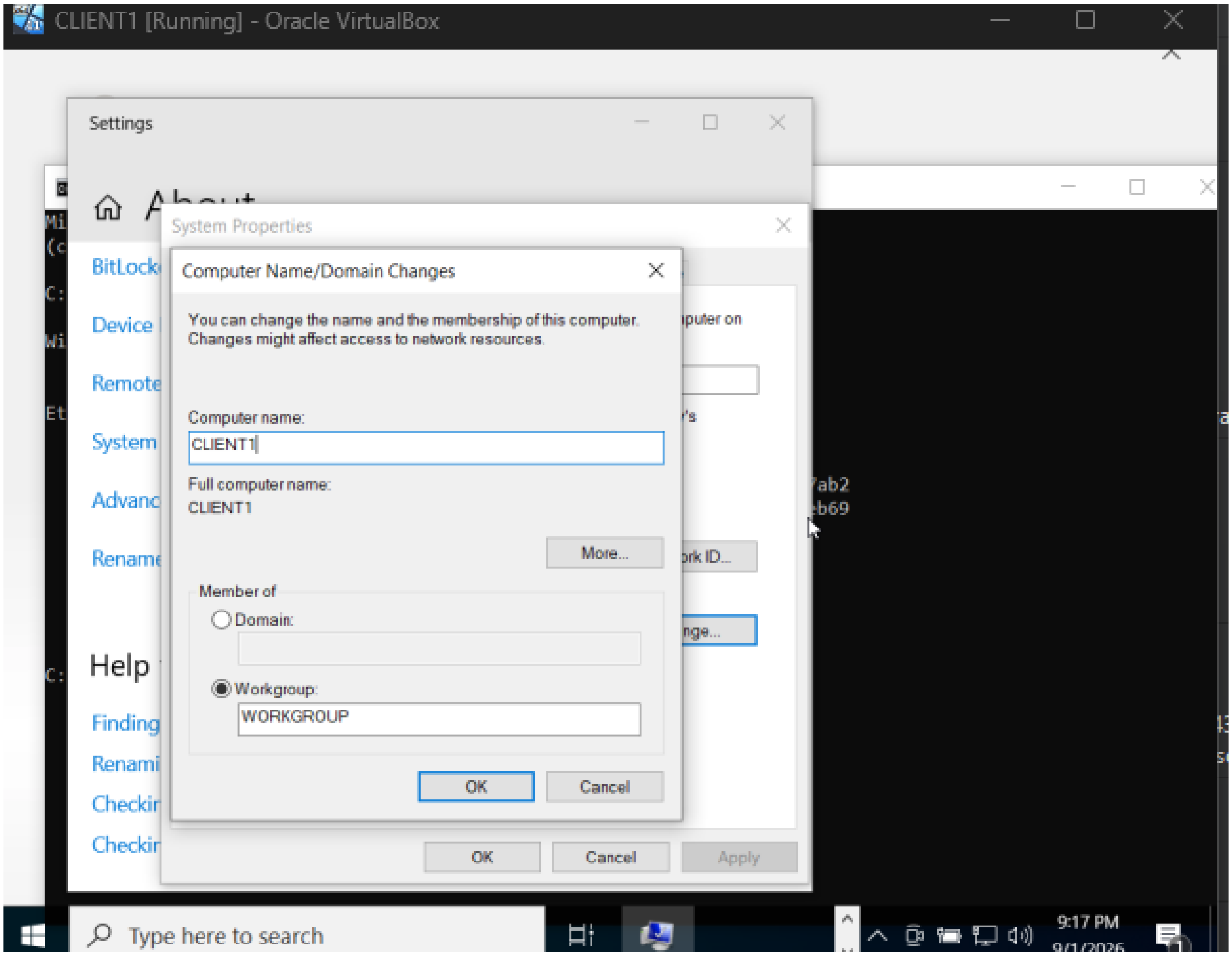

Figure 3. Configuring the Client Workstation

The client workstation is assigned the computer name CLIENT1.

At this stage, CLIENT1 remains a member of the default WORKGROUP and has not yet been joined to the enterprise domain.

Assigning the workstation a recognizable computer name provides a method of identifying the device as it is added and managed within the domain environment.

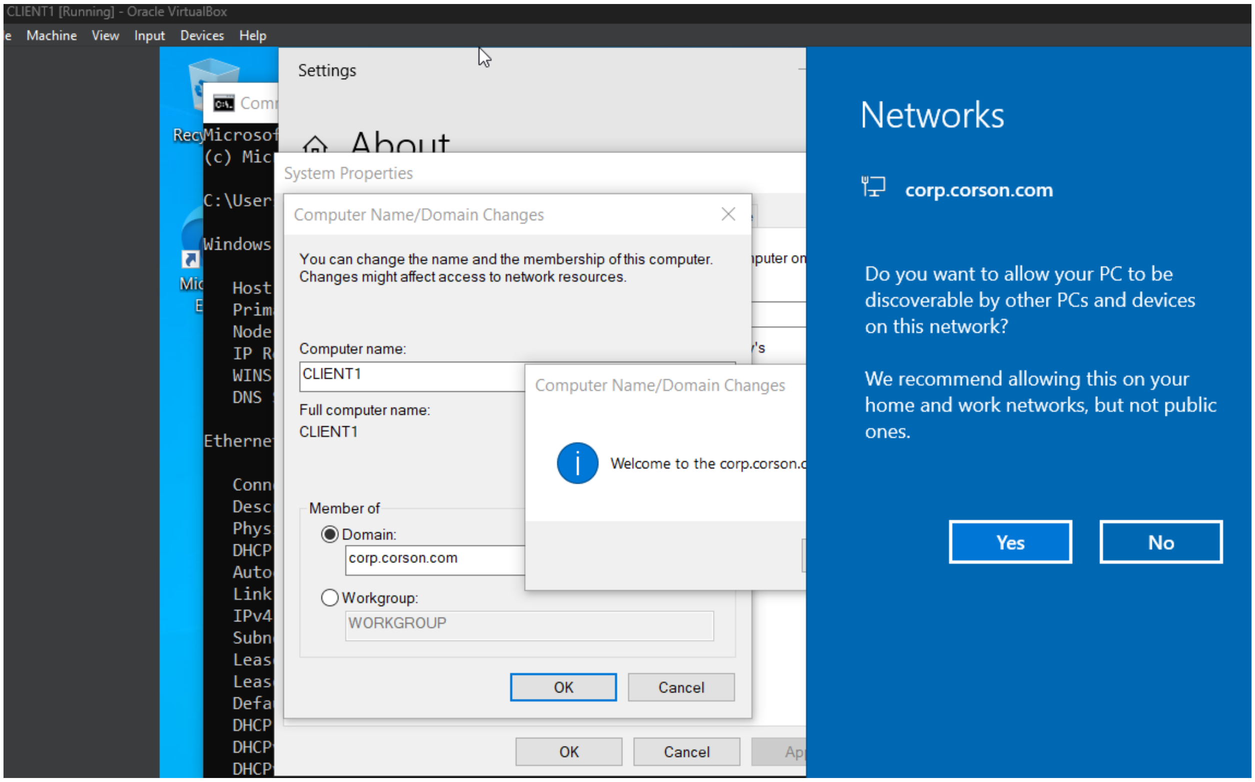

Figure 4. Joining CLIENT1 to the Enterprise Domain

CLIENT1 is configured to join the corp.corson.com domain. After the Domain Controller is successfully contacted and domain credentials are provided, the **"Welcome to the corp.corson.com domain"** message confirms that CLIENT1 successfully joined the Active Directory domain.

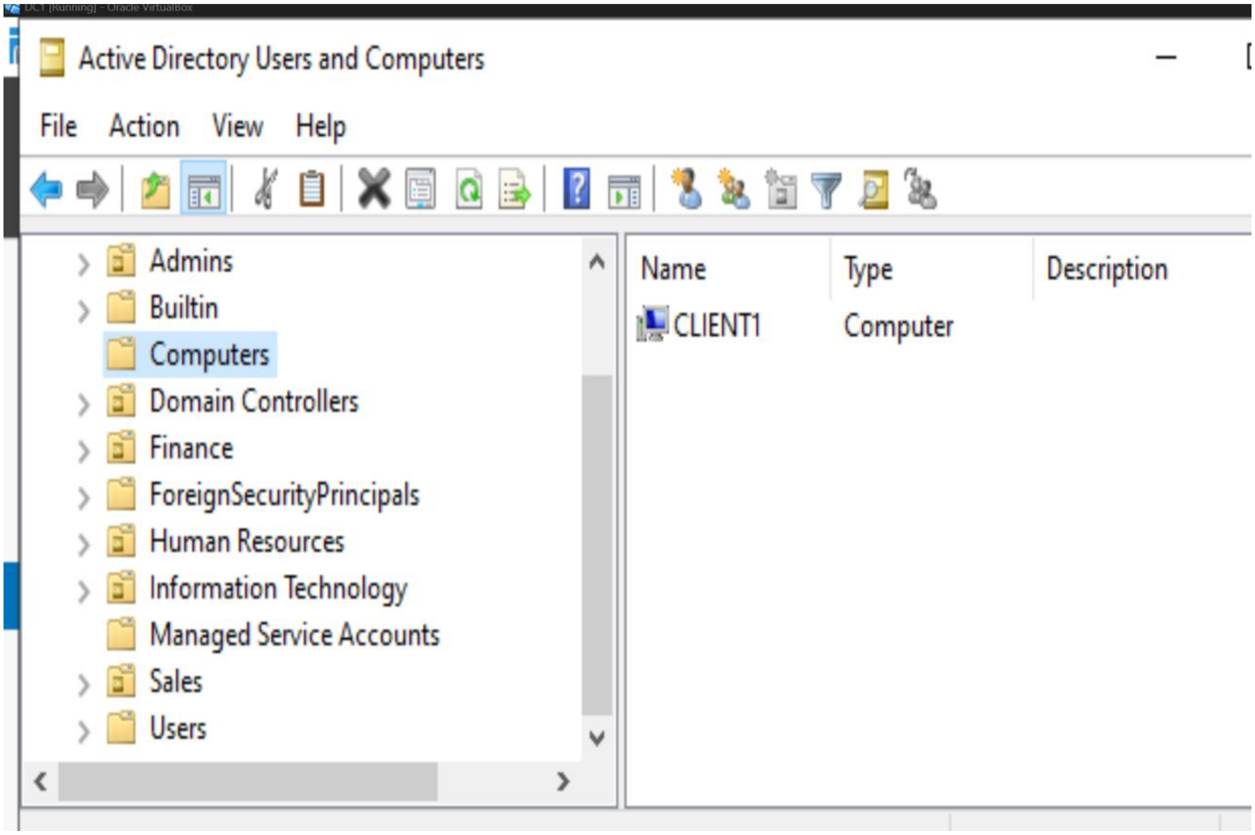

Figure 5. Verifying Client workstation on the AD

CLIENT1 is displayed as a Computer object within Active Directory Users and Computers after successfully joining the corp.corson.com domain. This verifies from the Domain Controller that the workstation has been added to and is now recognized as a domain-connected endpoint.

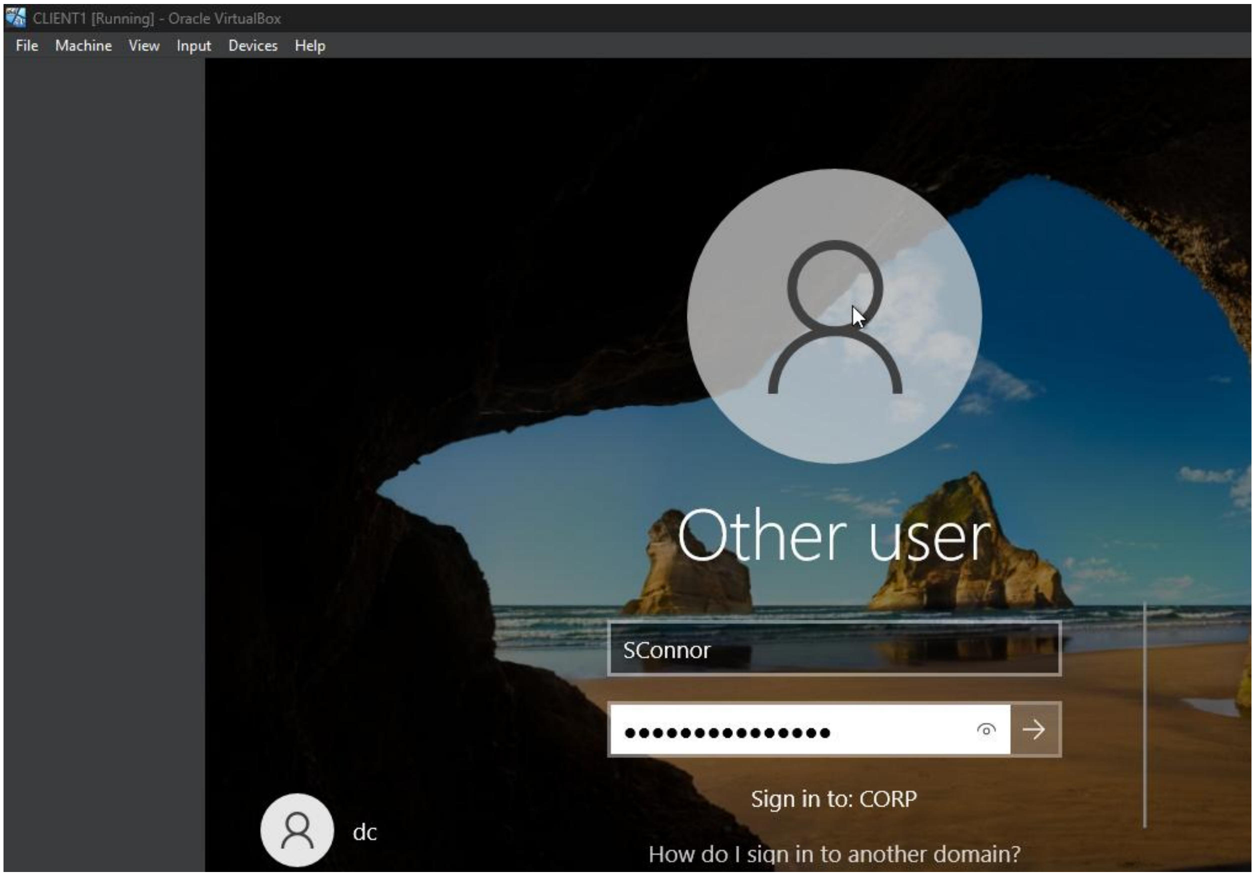

Figure 6. End user workstation login attempt

Following the domain join and system restart, the Windows sign-in screen provides the option to authenticate against the CORP domain. The enterprise account SConnor, previously created within Active Directory, is used to sign into the workstation.

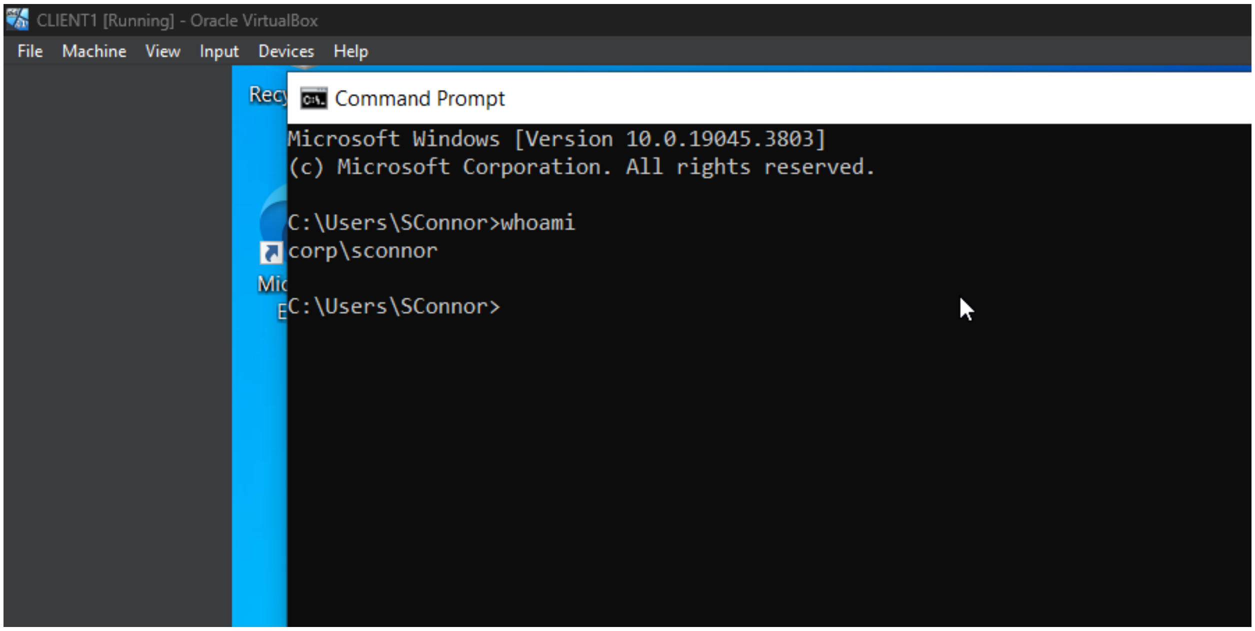

Figure 7. Verifying Enterprise User Authentication

After signing into CLIENT1 using the enterprise user account, the whoami command is executed through Command Prompt. The command returns: corp\sconnor This verifies that the current user session is authenticated using the SConnor account from the CORP domain rather than a local account stored on CLIENT1

What Changed?

Before this configuration, the corp.corson.com domain contained the server-side infrastructure required to support an enterprise environment, including Active Directory Domain Services, DNS, DHCP, Organizational Units, users, groups, shared resources, and permissions. However, a separate client workstation had not yet been connected to the environment.

A Windows 10 Pro virtual machine named CLIENT1 was created to represent an end-user workstation. CLIENT1 was connected to the internal domain network and received the IPv4 address 172.24.8.100 from the DHCP service configured on the Domain Controller. This verified that the DHCP configuration created earlier in the lab was successfully providing network configuration to a client.

CLIENT1 was then configured to join the corp.corson.com Active Directory domain. The successful domain join established the workstation as a domain connected endpoint capable of using services provided by the Domain Controller.

After joining the domain, the enterprise user account SConnor was used to sign into CLIENT1. The whoami command returned corp\sconnor, confirming that the user was successfully authenticated through the CORP domain rather than through a local workstation account.

This configuration connects the server side infrastructure created throughout the previous portions of the lab to an actual client workstation. The environment can now centrally authenticate enterprise users and support domain connected endpoints.